Inference for Paired Data

Presentation Outline

Variables and Data

Variables

- Two Similar Quantitative Variables
- Collected using same procedure with same units, etc.

Data Collection

- Random sample of size n from a given population
- Observations on each variable are paired or matched in some way
  - From same cases - Observations separated by time, space, etc.
  - From matched cases - Observations are from different cases, but cases are related.

Compare observations of two similar quantitative variables.

Examples of Paired Data

Ages of Heterosexual Partners

- Cases = Heterosexual Partners
- Variable 1 = Age of wife
- Variable 2 = Age of husband
- Observations of variable 1 and 2 are related since we know there is a tendency for people to be partnered with others of similar age.

Weight Loss after using GLP-1 medication for one year

- Cases = Weight Loss Study Participants
- Variable 1 = Weight prior to Study
- Variable 2 = Weight at completion of Study
- Observations of variable 1 and 2 are related. They come from the same person at two points in time.

Parameters for Quantitative Variables

Individual Variables

- μ_1 = mean of variable 1 in population
- μ_2 = mean of variable 2 in population

Given data collection, observations of two quantitative variables are related (i.e. not independent).

Define:

- $\mu_D = \mu_1 - \mu_2$ = mean difference between two values in population
- σ_D^2 = variance of difference between two values in population

Data

- $D_i = Y_{i1} - Y_{i2}$ = difference in value of variables for i^{th} case

Quantitative Variable 1	Quantitative Variable 2	Difference
Y_{11}	Y_{12}	$D_1 = Y_{11} - Y_{12}$
Y_{21}	Y_{22}	$D_2 = Y_{21} - Y_{22}$
Y_{31}	Y_{32}	$D_3 = Y_{31} - Y_{32}$
$\vdots$	$\vdots$	$\vdots$
Y_{n1}	Y_{n2}	$D_n = Y_{n1} - Y_{n2}$

Summary Statistics for Differences

Mean: $\overline{D} = \frac{\sum_{i=1}^{n} D_i}{n}$

Variance: $S_D^2 = \frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}$

Std. Dev.: $S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}}$

Null and Alternative Hypotheses

Null Hypothesis:

- There is no difference in the mean values of the two quantitative variables in the population.
- $H_0 : \mu_D = 0$

Alternative Hypothesis: Selection depends on study goals

- $H_a : \mu_D \neq 0$
- $H_a : \mu_D < 0$
- $H_a : \mu_D > 0$

Sampling Distribution of $\overline{D}$ under H_0

If H_0 is true, for large sample sizes n, we have:

$$z = \frac{\overline{D}}{\frac{\sigma_D}{\sqrt{n}}}$$

has a standard Normal distribution.

In order to use above result for inference, we would need to know σ_D, which is a population parameter and generally unknown.

Test Statistic

Estimating the unknown parameter σ_D with S_D gives:

$$t = \frac{\overline{D}}{\frac{S_D}{\sqrt{n}}}$$

For large sample sizes n, we have t has a t distribution with $n - 1$ degrees of freedom.

P-value

Large values (in magnitude) of t indicate evidence against the null hypothesis.

Quantify this evidence with p-value $=$ probability of obtaining the test statistic t or a value more extreme (in direction of H_a) if H_0 is true.

- $H_a : \mu_D \neq 0$
  p-value $= 2 * P(t_{n-1} > |t|)$
- $H_a : \mu_D < 0$
  p-value $= P(t_{n-1} < t)$
- $H_a : \mu_D > 0$
  p-value $= P(t_{n-1} > t)$

Example - Hypothesis Test for μ_D

Are any physiological indicators associated with schizophrenia? In a 1990 article, researchers reported the results of a study that controlled for genetic and socioeconomic differences by examining 15 pairs of monozygotic (identical) twins, where one of the twins was schizophrenic and the other was not. The researchers used magnetic resonance imaging to measure the volumes (in cm^3) of several regions and subregions of the twins' brains.

Variables:

- Unaffect = volume of left hippocampus of unaffected twin (in cm^3)
- Affected = volume of left hippocampus of affected twin (in cm^3)

Example - Hypothesis Test for μ_D

Data:

Unaffect	Affected	Difference
1.94	1.27	0.67
1.44	1.63	-0.19
1.56	1.47	0.09
⋮	⋮	⋮
2.08	1.97	0.11

Example - Hypothesis Test for μ_D

Null Hypothesis

- There is no mean difference in volumes of the left hippocampus between unaffected and affected twins in the population of identical twins.
- $H_0 : \mu_D = 0$

Alternative Hypothesis

- There is a mean difference in volumes of the left hippocampus between unaffected and affected twins in the population of identical twins.
- $H_0 : \mu_D \neq 0$

Example - Hypothesis Test for μ_D

Schizophrenia Data Sample Statistics

$$n = 15 \qquad \overline{D} = 0.199 \qquad S_D = 0.2383$$

Test Statistic

$$t = \frac{\overline{D}}{\frac{S_D}{\sqrt{n}}} = \frac{0.199}{\frac{0.2383}{\sqrt{15}}} = 3.2289$$

Example - Hypothesis Test for μ_D

p-value

$$2 * P(t_{14} > 3.2289) = 0.0061$$

Conclusion

We have very strong evidence of a mean difference in volumes of the left hippocampus between unaffected and affected twins in the population of identical twins.

Sampling Distribution for $\overline{D}$

For large sample sizes n, we have:

$$P\left(-t_{\alpha/2} < \frac{\overline{D} - \mu_D}{\frac{S_D}{\sqrt{n}}} < t_{\alpha/2}\right) = 1 - \alpha$$

where $t_{\alpha/2}$ is the upper $\alpha/2$ percentile from a t distribution with $n-1$ degrees of freedom.

Solving the above inequality for μ_D gives us:

Confidence Interval for μ_D

$$P\left(\overline{D} - t_{\alpha/2}\frac{S_D}{\sqrt{n}} < \mu_D < \overline{D} + t_{\alpha/2}\frac{S_D}{\sqrt{n}}\right) = 1 - \alpha$$

$(1-\alpha)\%$ Confidence Interval for μ_D

$$\overline{D} \pm t_{\alpha/2}\frac{S_D}{\sqrt{n}}$$

Example - Confidence Interval for μ_D

Sample Statistics

$$n = 15 \qquad \overline{D} = 0.199 \qquad S_D = 0.2383$$

95% Confidence Interval for μ_D

$$\overline{D} \pm t_{\alpha/2}\left(\frac{S_D}{\sqrt{n}}\right) = 0.199 \pm 2.1448\left(\frac{0.2383}{\sqrt{15}}\right) = (0.0667, 0.3306)$$

Example - Confidence Interval for μ_D

We are 95% confident the mean difference in volumes of the left hippocampus between unaffected and affected twins in the population of identical twins is between 0.0667 and 0.3306 cm^3.

Comparison to Two Sample Methods

Two Sample Methods

- Two Populations under Study
- Structure of Data Table
  - One Quantitative Variable
  - One Categorical Variable - indicates sample/population membership
- Goal: Compare Mean of Quantitative Variable between Populations

Paired Data

- One Population under Study
- Structure of Data Table
  - Two Similar Quantitative Variables
- Goal: Determine Mean Difference between related values of two quantitative variables.